Week 6 Notes

The Inverse of the Exponential Function: Logarithms, Exponential Equations and the Laws of Logarithms

Course: Mathematics–1

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Lecture 1 — Inverse of the Exponential Function

Logarithmic functions; the correspondence rule; domains; graphs by reflection

1 Recalling the exponential function

In the previous lecture we saw the inverse of a general function, and concluded that if a function is one-to-one then finding its inverse is very easy. Let us now focus on the inverse of the **exponential function** — its properties, how it is graphed, and the domain and range of the resulting inverse.

Recall our notion of the exponential function. A function is called an exponential function if it can be written in the form

$$f(x) = a^x, \qquad a > 0, \quad a \neq 1.$$

Why these conditions on a

- $a > 0$ is needed because otherwise we would have to deal with complex variables, which is out of scope for this course.
- $a \neq 1$ is needed because putting $a = 1$ gives $f(x) = 1^x = 1$ for all x — a constant function, which is not an interesting function to study.

Whenever these conditions are enforced, we know that $f(x) = a^x$ is **one-to-one**. And since every one-to-one function has an inverse, this function also has an inverse — there is nothing special about it. That inverse is what we define as the **logarithmic function**. Naturally, since we are talking about an exponential function with base a, we obtain a logarithmic function with base a.

2 Definition of the logarithmic function

Definition 1.1 — Logarithmic function to the base a

The logarithmic function to the base a, in the standard form, is given by

$$\boxed{y = \log_a x}$$

and is defined to be the **inverse** of the function $f(x) = a^x$. That is,

$$f^{-1}(x) = \log_a x.$$

Here $\log_a$ denotes the function, x is the argument of the function, and the value is plotted along the y-axis. It is simply the function f^{-1} written explicitly.

3 The correspondence between log and exponential

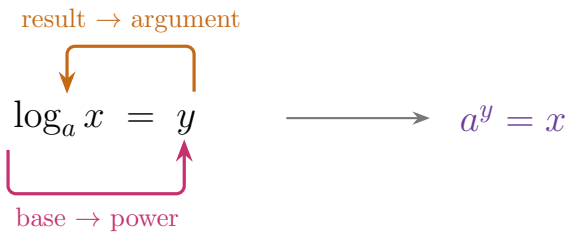
To keep track of exactly what is happening, and how the logarithmic function relates to the exponential function, we use the following one-to-one correspondence.

$$y = \log_a x \qquad \Longleftrightarrow \qquad x = a^y$$

3.1 The "7 rule" — a way to remember it

Write the statement as $\log_a x = y$. Then you may virtually assume the following: start from the **base**, go across to the **right-hand side**, and come back to the **argument**. That is, start with a, go to the right-hand side which becomes the **power**, and that should give you x:

$$a^y = x.$$



This is a simple technique for memorising the one-to-one correspondence between the logarithm and the exponential. You can easily see that if $x = a^y$ and we want to know the value of y, we should be able to get it by taking the logarithm of x.

4 Verifying that $\log_a$ really is the inverse

We defined $\log_a$ to be the inverse of f; but is it actually the inverse? As stated earlier, we check the two rules

$$f\big(f^{-1}(x)\big) = x \qquad \text{and} \qquad f^{-1}\big(f(x)\big) = x.$$

Check 1: $f\big(f^{-1}(x)\big)$

Since $f^{-1}(x) = \log_a x$ and f raises a to its argument,

$$f\big(f^{-1}(x)\big) = a^{\,f^{-1}(x)} = a^{\log_a x} = x,$$

using the one-to-one correspondence above.

Check 2: $f^{-1}\big(f(x)\big)$

Similarly,

$$f^{-1}\big(f(x)\big) = \log_a\big(f(x)\big) = \log_a\left(a^x\right) = x.$$

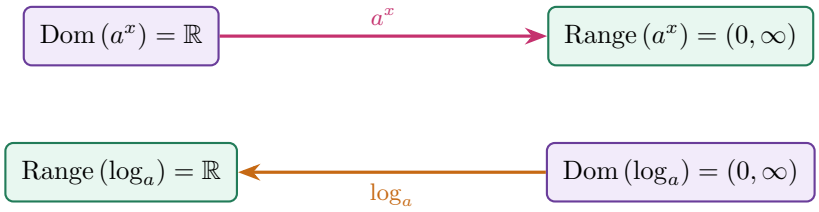
5 Domain and range

To understand this completely we need the domain and range of both functions. We already know, from our study of exponential functions, that

$$\text{Dom}\,(a^x) = \mathbb{R}, \qquad \text{Range}\,(a^x) = (0, \infty),$$

since x can be the entire real line, while a^x can never take negative values.

If the logarithmic function is to be defined as the inverse, then these two must swap over — this is the crux of the definition of an inverse:



Domains and ranges to remember

$$\text{Dom}\,(\log_a) = \text{Range}\,(a^x) = (0, \infty), \qquad \text{Range}\,(\log_a) = \text{Dom}\,(a^x) = \mathbb{R}.$$

So the logarithmic function is defined as

$$\log_a : (0, \infty) \longrightarrow \mathbb{R}.$$

In the domain it cannot take negative values, and it cannot take 0 either; in the range it covers the entire real line.

These two points are important, as they help in working out the domains of any functions *derived* from logarithmic or exponential functions. Always remember the valid domains and ranges. This also completes the verification that the logarithm, as we have defined it, really is the inverse of the exponential function.

6 Domains of derived functions

By derived functions we mean compositions built from the basic logarithmic function.

6.1 Example 1

Example 1.1

Find the domain of

$$f(x) = \log_4(1 - x).$$

The function $\log_4$ has domain $(0, \infty)$, which means the argument supplied to it cannot be 0, nor can it be negative. Looking at the argument $1 - x$, we therefore require

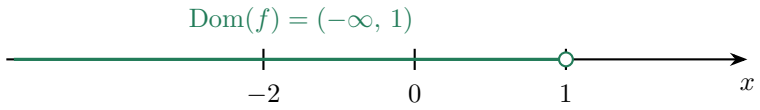
$$1 - x > 0 \qquad \Longleftrightarrow \qquad 1 > x.$$

Can x be less than 0? Yes — if $x < 0$ then $1 - x$ is certainly greater than 0, so negative values of x cause no trouble at all. The only condition we need is $x < 1$. Hence

$$\text{Dom}(f) = (-\infty, 1).$$

A common mistake

The answer is **not** $(1, \infty)$. This is exactly the kind of slip that is easy to make — read the inequality carefully: $1 - x > 0$ means x is *less* than 1, so the domain runs from $-\infty$ up to 1 and cannot go beyond 1.



6.2 Example 2

Example 1.2

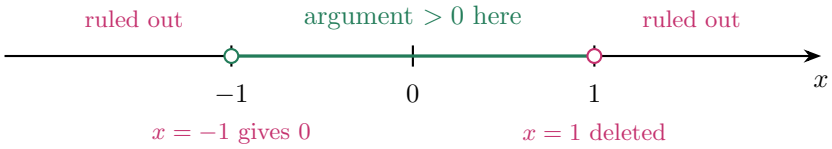
Find the domain of

$$g(x) = \log_3\left(\frac{1+x}{1-x}\right), \qquad x \neq 1.$$

First, $\log_3$ is defined only when the argument lies in $(0, \infty)$. So we need the argument trapped in that interval:

$$\frac{1+x}{1-x} > 0.$$

When can this happen? Let us split the real line into parts, marking the points -1, 0 and 1.



The reasoning goes as follows:

- $x = 1$ is already excluded — that point is deleted, since the denominator vanishes there.
- If $1 - x > 0$, i.e. $x < 1$, the denominator is positive. But we must not simply take $(-\infty, 1)$, because of the numerator $1 + x$.
- The numerator $1 + x$ becomes negative when $x < -1$, and for such x the denominator $1 - x$ is positive — so the whole fraction is negative. That part must be ruled out: $(-\infty, -1)$ is excluded.
- At $x = -1$ the numerator is 0, so the argument is 0, which is also not allowed. So $x = -1$ is ruled out too.

We are therefore left only with the interval $(-1, 1)$:

$$\mathrm{Dom}(g) = (-1, 1).$$

Why the value zero is fine

You might ask why 0 is not a problem. Substituting $x = 0$ gives

$$g(0) = \log_3\left(\frac{1+0}{1-0}\right) = \log_3 1,$$

and the argument 1 is a perfectly valid positive number, so the function is well defined there.

Before trying to solve any problem involving logarithms, we must first verify whether the problem is defined on a proper domain. Most of the time, when a problem is formulated, it may not be defined on a proper domain — and then solving it is a meaningless exercise. So always ensure your problem is defined on a valid domain.

7 Reversibility of log and exponential

Let us now take an example that demonstrates the reversibility of the logarithm.

Take

$$y = \log_3 x, \qquad x \in (0, \infty),$$

so that y belongs to the real line. If we now write 3^y, then

$$3^y = 3^{\log_3 x} = x,$$

by definition, since the exponential is the inverse of the log. Therefore $3^y = x$.

A fact we use repeatedly

$$a^u = a^v \quad \text{for } a > 0,\ a \neq 1 \qquad \Longrightarrow \qquad u = v.$$

7.1 Example: evaluating $\log_3\left(\frac{1}{9}\right)$

Example 1.3

Find $\log_3\left(\frac{1}{9}\right)$.

Look at the 9 and the 3. We know

$$3^2 = 9 \qquad \Longrightarrow \qquad 3^{-2} = \frac{1}{9}.$$

So substitute for the argument:

$$\log_3\left(\frac{1}{9}\right) = \log_3\left(3^{-2}\right).$$

But $\log_a\left(a^x\right) = x$, and therefore

$$\log_3\left(\frac{1}{9}\right) = -2.$$

This is how such problems can be solved very easily whenever the argument can be identified as a power of the base.

8 Graph of the logarithmic function

We have identified the inverse of the exponential function: it is the logarithmic function to the *same* base. Let us now look at the graph of

$$f(x) = \log_a x.$$

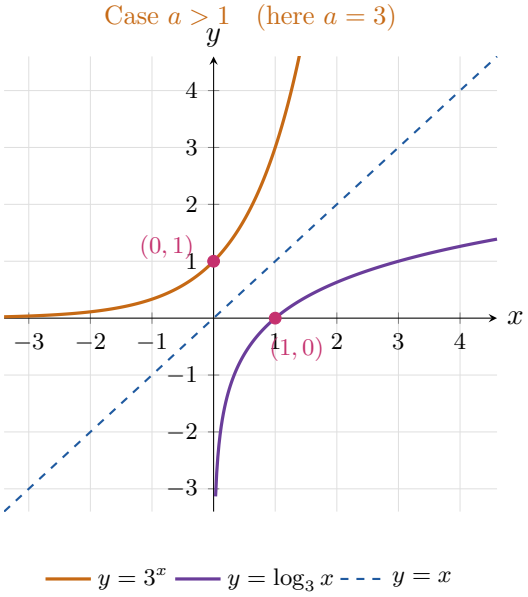
Recall that the graphs of exponential functions had two cases, split at $a = 1$: one picture for $0 < a < 1$ and a different picture for $a > 1$. Recall also, from the previous lecture, the interpretation of the graph of an inverse function:

Graph of an inverse function

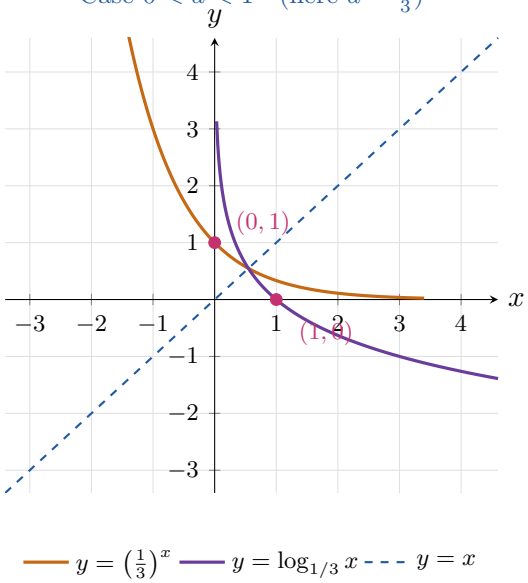
The graph of the inverse function is the **reflection of the original function f in the line $y = x$** — its mirror image.

So we take a point such as $(0, 1)$ on the exponential graph, and its reflection $(1, 0)$ lies on the logarithmic graph. Continuing to translate points this way and joining them up gives the whole curve. All of this is possible because our function is one-to-one.

Case $a > 1$ (here $a = 3$)



Case $0 < a < 1$ (here $a = \frac{1}{3}$)



In the case $a > 1$ there are no overlaps, so the reflection is relatively easy to draw: each point is carried across the line $y = x$, and joining them gives the exact mirror image of the original curve, running close to it on the other side of $y = x$. In the case $0 < a < 1$ the same procedure applies — the point $(0, 1)$ goes to $(1, 0)$, points in the upper region are carried to the right, and points further along are carried into the lower region, giving a curve that is asymptotic to the y-axis.

Seeing all four graphs together makes it clear how easy it is to draw the graph of an inverse function once we know the graph of the original.

In the next lecture we will use our knowledge of logarithmic functions to see how the formulation of mathematical problems becomes easier with logarithms — even though the logarithmic function carries the limitation of being defined only on $(0, \infty)$ and not on the whole real line.

Lecture 2 — Properties and Graphs of Logarithmic Functions

Graphing by transformations; the natural logarithm $\ln x$ and the common logarithm

1 Recalling the two graphs

In the last lecture we saw how to graph $f(x) = \log_a x$, and found two kinds of graphs — one form when $0 < a < 1$, and another form when $a > 1$.

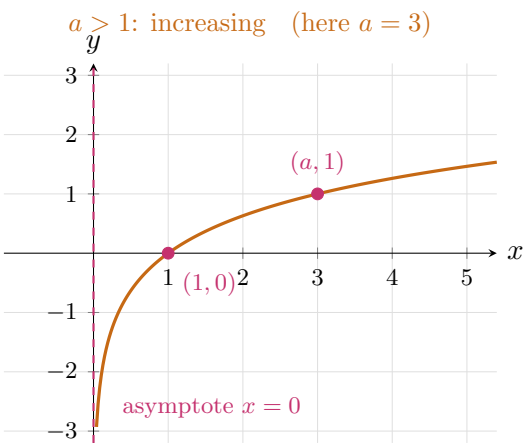
In particular: for $0 < a < 1$, take the graph of a^x and reflect it in the line $y = x$ to get the graph of $\log_a x$. For $a > 1$ matters are even easier, since there is no intersection to worry about — simply reflect the exponential curve in the line $y = x$ to get the logarithmic curve. Let us now list all the properties of the graph of the logarithmic function.

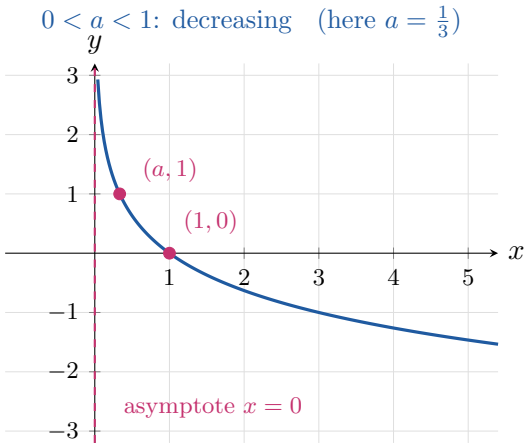
2 Properties of the graph of $\log_a x$

Property	$f(x) = a^x$	$f(x) = \log_a x$
Domain	$\mathbb{R}$	$(0, \infty)$
Range	$(0, \infty)$	$\mathbb{R}$
x-intercept	None	$(1, 0)$
y-intercept	$(0, 1)$	None
Asymptote	Horizontal: $y = 0$	Vertical: $x = 0$
Passes through	$(0, 1)$ and $(1, a)$	$(1, 0)$ and $(a, 1)$
One-to-one	Yes	Yes
For $0 < a < 1$	Decreasing	Decreasing
For $a > 1$	Increasing	Increasing

A few remarks on the entries:

- The exponential function had the y-intercept $(0, 1)$ and no x-intercept. For the logarithm these swap over: the x-intercept is $(1, 0)$ and there is **no y-intercept** — simply because the graph is a reflection in the line $y = x$.
- Where the exponential had a **horizontal** asymptote (the x-axis), the logarithm has a **vertical** asymptote, namely $x = 0$. For $a > 1$ the curve runs down the negative side of the y-axis; for $0 < a < 1$ it runs up the positive side.
- Since the exponential passed through $(0, 1)$ and $(1, a)$, the logarithm always passes through the two static points $(1, 0)$ and $(a, 1)$.





While drawing the graph of a logarithmic function in standard form, always check whether your sketch satisfies these properties — it is a good cross-check on whether your answer is correct.

3 Example 1: $f(x) = -\log_4(x+1)$

Example 2.1

Draw the graph of

$$f(x) = -\log_4(x+1).$$

We build the graph in three stages, remembering the domain at every step.

Stage 1: the graph of $\log_4 x$

First ask: is the base greater than 1 or less than 1? Here the base is $4 > 1$, so the function is **increasing**. It is one-to-one, the curve must pass through

$$(1, 0) \qquad \text{and} \qquad (a, 1) = (4, 1),$$

and there is a vertical asymptote at $x = 0$. Being increasing, the curve comes up from below. Also

$$\text{Dom} = (0, \infty), \qquad \text{Range} = \mathbb{R}.$$

Stage 2: the shift to $\log_4(x+1)$

We are not given $\log_4 x$ but $\log_4(x+1)$. Whatever value x was taking, now $x+1$ takes it — this is a **shift of 1 unit along the x-axis, to the left**. Everything translates by one unit:

$$\text{asymptote } x = 0 \longrightarrow x = -1, \qquad (1, 0) \longrightarrow (0, 0), \qquad (4, 1) \longrightarrow (3, 1).$$

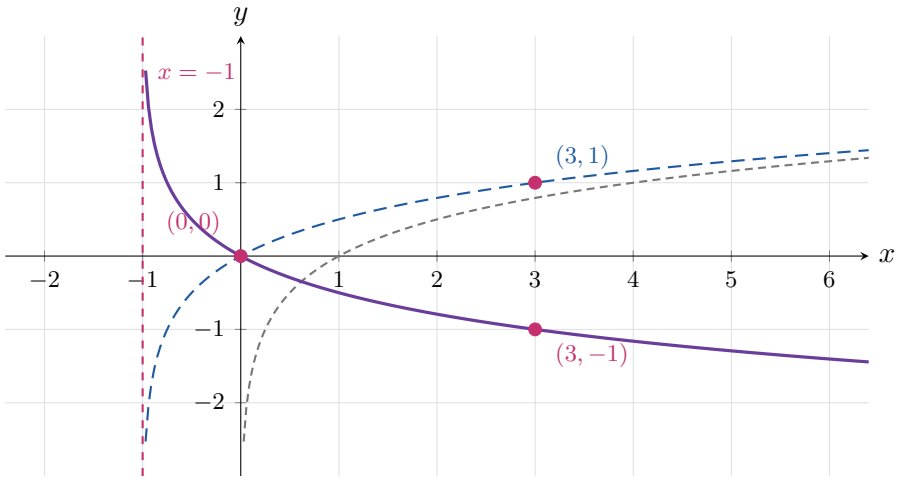
The values on the y-axis do not change; only the values on the x-axis change.

Stage 3: the minus sign

Now comes the twist — the minus sign. Writing $y = \log_4(x+1)$, attaching a minus sign replaces y by $-y$, which is a **reflection in the x-axis**. The upper side goes down and the lower side comes up:

$$(0, 0) \longrightarrow (0, 0) \quad \text{(unchanged)}, \qquad (3, 1) \longrightarrow (3, -1).$$

The asymptote $x = -1$ stays where it is.



Stage 1: $\log_4 x$ — Stage 2: $\log_4(x+1)$ — Stage 3: $-\log_4(x+1)$

4 Example 2: $g(x) = \log_{1/4}(-x) + 1$

Example 2.2

Draw the graph of

$$g(x) = \log_{1/4}(-x) + 1.$$

Stage 1: the graph of $\log_{1/4} x$

Can we draw $\log_{1/4} x$? Yes. Here the base is $\frac{1}{4} < 1$, so the function is **decreasing**, with vertical asymptote $x = 0$, and it passes through

$$(1, 0) \qquad \text{and} \qquad (a, 1) = \left(\tfrac{1}{4},\, 1\right).$$

Stage 2: replacing x by $-x$

The twist is that the argument is $-x$. Wherever x appeared, we replace it by $-x$, which simply changes the paradigm — we take a **reflection in the y-axis**. The line $x = 0$ still remains the asymptote, and the graph flips over to become the exact mirror image:

$$(1, 0) \longrightarrow (-1, 0), \qquad \left(\tfrac{1}{4}, 1\right) \longrightarrow \left(-\tfrac{1}{4}, 1\right).$$

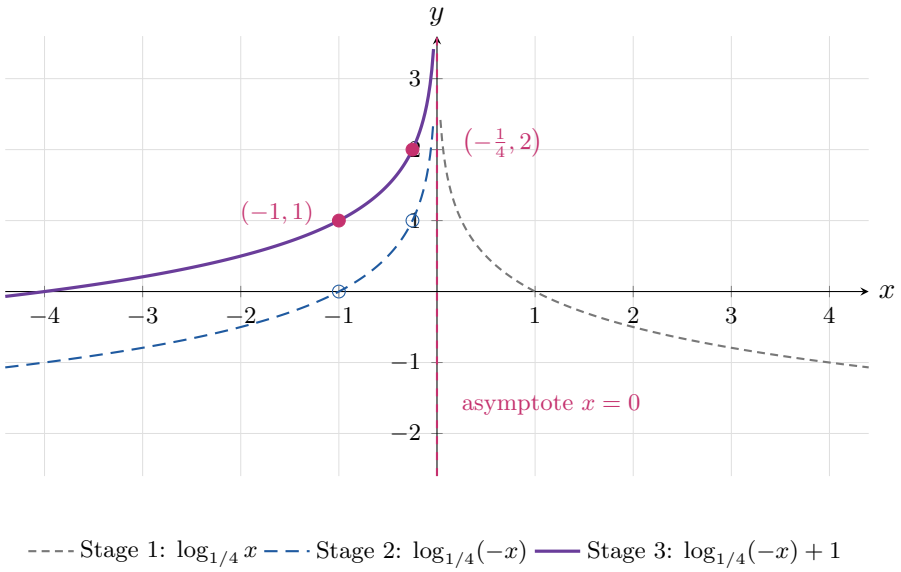
Note that the domain is now $-x > 0$, that is $x < 0$.

Stage 3: adding 1

Finally we add 1. Earlier we had $y = \log_{1/4}(-x)$; adding 1 simply **shifts all values one level up along the y-axis**:

$$(-1, 0) \longrightarrow (-1, 1), \qquad \left(-\tfrac{1}{4}, 1\right) \longrightarrow \left(-\tfrac{1}{4}, 2\right).$$

We are able to map those two points, so the verification of the graph is complete.



5 The natural logarithmic function

Alongside the logarithmic function to a general base a, there is one special logarithm — the **natural logarithmic function** — which involves Euler's constant e. It is special in exactly the way the natural exponential function is special, and so it is defined separately.

Definition 2.1 — Natural logarithm

The **natural logarithmic function** is the logarithm to the base e,

$$\log_e x,$$

and it has its own special notation:

$$\boxed{\ln x \;=\; \log_e x}$$

Here l stands for *logarithm* and n stands for the *natural* base.

Whenever we write $\ln x$, it simply means we are talking about the natural logarithm of x, that is, $\log_e x$. Henceforth, whenever we discuss the natural logarithmic function, we use this standard notation.

The verification is simple, and is just the inverse relationship again:

$$\ln\left(e^x\right) = x \quad \text{for } x \in \mathbb{R}\ \big(\text{the domain of } e^x\big), \qquad e^{\ln x} = x \quad \text{for } x \in (0, \infty)\ \big(\text{the domain of } \ln x\big).$$

6 The common logarithm

In a similar manner there is another special logarithm. While the one above is called the natural logarithm, this one is called the **common logarithm**, and it has to do with our decimal representation. It is denoted by log written *without any base*:

$$\log_{10} x \;=\; \log x.$$

Reading the notation

In common terminology, when no base is mentioned you may take the logarithm to be to the base 10; and when something like $\ln x$ is written, that is $\log_e x$.

In the old days of calculators there were two separate keys for this: one key marked ln and another marked log. The log key referred to base 10, and the ln key to the natural base. So remember these two commonly used logarithms, which we will use quite often in daily practice — calculators distinguish them as ln and log, and we prefer that you distinguish them the same way.

Name	Base	Notation
Natural logarithm	e	$\ln x = \log_e x$
Common logarithm	10	$\log x = \log_{10} x$
General logarithm	$a > 0,\ a \neq 1$	$\log_a x$

There is nothing else special about the natural logarithm — it is just a way of taking logarithms with e as the base.

Lecture 3 — Solving Exponential Equations

Using logarithms; quadratic-in-form equations; unequal bases; graphical solutions

1 The property we keep using

Let us now go back and solve exponential equations — this is where logarithms come in handy. The single property we track throughout is

$$\boxed{\log_a\left(a^x\right) = x}$$

Everything below is a matter of arranging the equation so that this property can be applied. Recall also that the logarithm is a **one-to-one** function, so hitting both sides of an equation with a logarithm changes nothing about the solutions.

2 Example 1: matching powers of the same base

Example 3.1

Solve for x:

$$2^{\,x+1} = 64.$$

This is not so much about logarithms as about **inspection** — 64 is a nice number. Write

$$64 = 16 \times 4 = 2^4 \times 2^2 = 2^6.$$

So the equation becomes

$$2^{\,x+1} = 2^6 \qquad \Longleftrightarrow \qquad 2^x = 2^5,$$

taking one factor of 2 out from the left. Now hit both sides with the logarithm to the base $a = 2$:

$$\log_2\left(2^x\right) = \log_2\left(2^5\right).$$

Using the property $\log_a\left(a^x\right) = x$,

$$\boxed{x = 5}$$

3 Example 2: reducing to a quadratic

Example 3.2

Solve for x:

$$e^{-x^2} = \frac{\left(e^x\right)^2}{e^3}.$$

This is also an exercise in computation, but it becomes slightly more complicated because it involves a term containing x^2. First simplify the right-hand side using the laws of indices:

$$\left(e^x\right)^2 = e^{2x}, \qquad \frac{1}{e^3} = e^{-3} \qquad \Longrightarrow \qquad \frac{\left(e^x\right)^2}{e^3} = e^{\,2x-3}.$$

So the equation is

$$e^{-x^2} = e^{\,2x-3}.$$

Take the natural logarithm of both sides and use the same property:

$$-x^2 = 2x - 3.$$

That simply means we have a **quadratic equation**:

$$x^2 + 2x - 3 = 0.$$

Use our knowledge of quadratic equations — this one is easily solved by **factoring**. Split the middle term:

$$x^2 + 3x - x - 3 = 0 \implies x(x+3) - 1(x+3) = 0 \implies (x+3)(x-1) = 0.$$

Hence

$$\boxed{x = -3 \quad \text{and} \quad x = 1}$$

Always check feasibility

Remember that we are solving equations where the domains and co-domains may not be properly defined, in which case we can land up with an **infeasible solution**. So always check whether the values actually work. Substituting $x = -3$ and $x = 1$ back into the original equation, both work perfectly — so both are genuine solutions to the problem.

4 Example 3: a quadratic-in-form equation

Example 3.3

Solve for x:

$$9^x - 2 \cdot 3^{x+1} - 27 = 0.$$

Before handling the equation, look closely: there is a common feature between the two exponents. Since $3^2 = 9$,

$$9^x = (3^x)^2,$$

and taking the $+1$ out of the second term,

$$2 \cdot 3^{x+1} = 2 \cdot 3 \cdot 3^x = 6 \cdot 3^x.$$

Using this trick, the equation becomes

$$(3^x)^2 - 6 \cdot 3^x - 27 = 0.$$

This is very similar to a quadratic equation — we may call it a **quadratic-in-form** equation. We can solve it for 3^x, not directly for x. Put

$$t = 3^x \qquad \Longrightarrow \qquad t^2 - 6t - 27 = 0.$$

Again resort to factoring, using $27 = 9 \times 3$:

$$t^2 - 9t + 3t - 27 = 0 \qquad \Longleftrightarrow \qquad (t-9)(t+3) = 0.$$

Now re-substitute $t = 3^x$:

$$(3^x - 9)\,(3^x + 3) = 0 \qquad \Longrightarrow \qquad 3^x = 9 \quad \text{or} \quad 3^x = -3.$$

An alarm should go off here

The possibility $3^x = -3$ is **not possible for any real** x, since an exponential function is always positive. That option is **infeasible** and must be discarded.

For the remaining case, hit it with a logarithm — or note more obviously that $9 = 3^2$:

$$3^x = 9 = 3^2 \qquad \Longrightarrow \qquad \boxed{x = 2}$$

Verification

Substitute $x = 2$ into the original equation as a cross-check:

$$9^2 = 81, \qquad 3^{2+1} = 3^3 = 27, \qquad 2 \times 27 = 54, \qquad 54 + 27 = 81. \checkmark$$

So the equation is satisfied and the problem is solved successfully — but remember the occurrence of the infeasible solution along the way.

5 Example 4: when the bases are not the same

Example 3.4

Solve for x:

$$5^{x-2} = 3^{3x+2}.$$

Unlike the previous problems, here the bases are **not the same**. In this case we rely on the logarithm and simply hit both sides — with the natural logarithm, say (the common logarithm would do just as well):

$$\ln\left(5^{x-2}\right) = \ln\left(3^{3x+2}\right).$$

What hitting with a log does

If you hit with the log of the *same* base, the base vanishes. If you hit with the log of some *other* base, that number remains unperturbed — but the exponent comes down in front. So here the exponents $x - 2$ and $3x + 2$ come out in front:

$$(x-2)\ln 5 = (3x+2)\ln 3.$$

Now $\ln 5$ and $\ln 3$ are merely *numbers* multiplying x and the constants — they are just constants. So our equation is in fact a **linear equation**. Collect the terms in x on one side and the constants on the other:

$$-2\ln 5 - 2\ln 3 = 3x\ln 3 - x\ln 5.$$

Take x common on the right:

$$-2\big(\ln 5 + \ln 3\big) = x\big(3\ln 3 - \ln 5\big).$$

As we will learn shortly, $\ln 5 + \ln 3 = \ln 15$, so

$$-2\ln 15 = x\big(3\ln 3 - \ln 5\big).$$

The expression in the bracket is not equal to 0, so we may divide by it. Noting also that $3\ln 3 = \ln 27$,

$$x = \frac{-2\ln 15}{3\ln 3 - \ln 5} = \frac{-2\ln 15}{\ln 27 - \ln 5}.$$

This can be simplified further: $-2\ln 15 = \ln\left(15^{-2}\right) = \ln\left(\frac{1}{225}\right)$ since $15^2 = 225$, and $\ln 27 - \ln 5 = \ln\left(\frac{27}{5}\right)$. Therefore

$$\boxed{x = \frac{\ln\left(\dfrac{1}{225}\right)}{\ln\left(\dfrac{27}{5}\right)}}$$

This is again just a number, and it resolves the problem for x. Is it feasible? Yes — these numbers are perfectly feasible, so there is no violation here. (How to carry out such simplifications in general is something we take up a little later.)

6 Example 5: when only a graphical solution is possible

Example 3.5

Solve for x:

$$x + e^x = 2.$$

Before hitting anything with a logarithm, note that we are facing a **plus sign**, so first rewrite the equation as

$$e^x = 2 - x.$$

Now hit it with the natural logarithm:

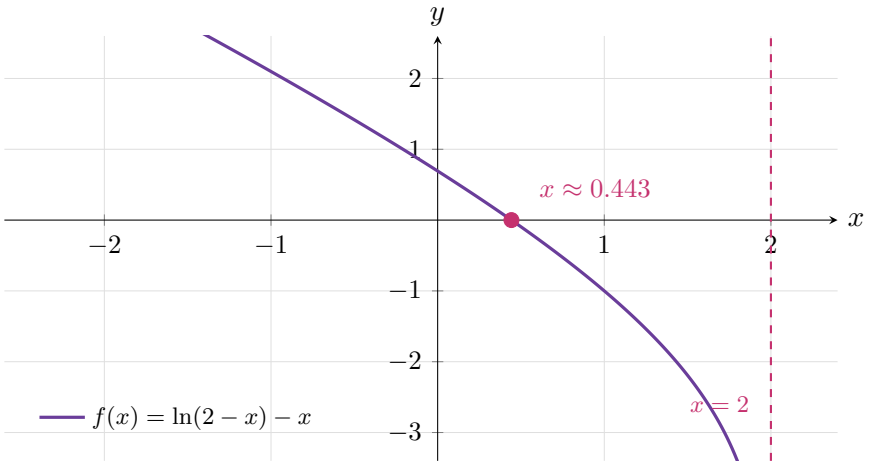
$$x = \ln(2 - x),$$

and here we are stuck — we cannot go anywhere from this point. If we try to hit it with the exponential again, we simply return to the same equation, so that is of no use.

So let us instead **graph** the equation. Consider the function

$$f(x) = \ln(2 - x) - x = 0,$$

and plot it with a graphical tool such as **Desmos**. The point where the curve cuts the horizontal axis is the solution.

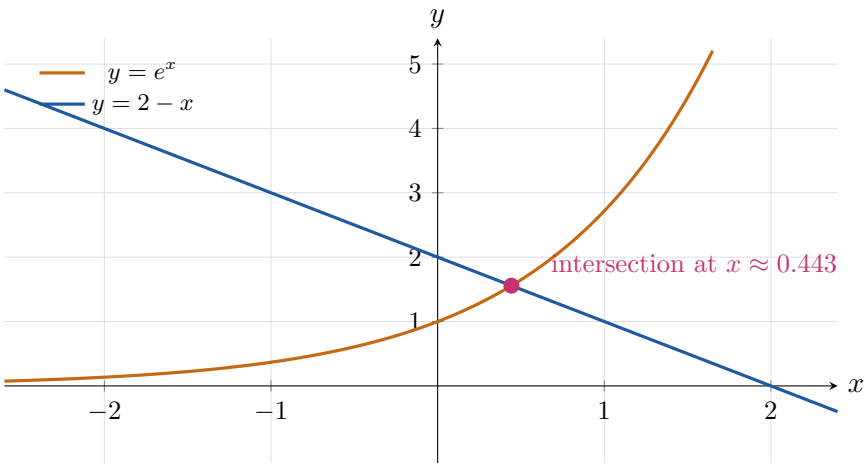


Checking with Desmos, the crossing point is at

$$\boxed{x \approx 0.443}$$

which is the approximate solution to this problem — and in such a problem this is the best that can be done.

Equivalently, one may plot the two sides of $e^x = 2 - x$ separately and look for their point of intersection, which occurs at the same value of x.



So we have now seen how to solve exponential problems both by **algebraic methods** and, when those run out, by **graphical methods**.

Lecture 4 — Laws of Logarithms

Basic properties; the product, quotient, reciprocal and power laws, with proofs

1 Properties we already have

Let us look at further properties of logarithmic functions. Throughout, we work under the standard set of conditions imposed on the base:

$$a \in (0,1) \qquad \text{or} \qquad a > 1.$$

We already know a few properties. First, the logarithmic function is the **inverse** of the exponential function, which is conveyed through the pair

$$a^{\log_a x} = x, \qquad \log_a (a^x) = x.$$

Next, from the graphical representation of the logarithm:

- $\log_a 1 = 0$ — that is, the point $(1,0)$ lies on the graph of the logarithmic function, independently of whether $a < 1$ or $a > 1$.
- $\log_a a = 1$ — that is, the point $(a,1)$ lies on the graph.

So for the logarithmic function the two reference points are $(1,0)$ and $(a,1)$; for the exponential function, being the reflection in the line $y = x$, the corresponding points are $(0,1)$ and $(1,a)$.

2 Two quick demonstrations

Since $a^{\log_a x} = x$, complicated-looking numbers can simplify at once.

Example 4.1

Simplify $3^{\log_3(\pi/2)}$.

Here $a = 3$ and the argument is $\frac{\pi}{2}$, so applying the property directly,

$$3^{\log_3(\pi/2)} = \frac{\pi}{2}.$$

Example 4.2

Simplify $4^{\log_4 1}$.

We already know $\log_4 1 = 0$, so

$$4^{\log_4 1} = 4^0 = 1.$$

Simple tricks like these are worth practising on many problems, because applying a logarithm or an exponential at the right moment plays a crucial role while solving problems on logarithmic and exponential functions.

3 The laws of logarithms

Why are these called *laws*? Because they are the very principles for which logarithms were invented by **Napier**. Astronomers needed to multiply distances of the order of 10^{32}, and multiplying such numbers by hand is a tedious task. Logarithms were invented to handle exactly this — in an era with no computational power at all — and so these results are honoured as the laws of logarithms.

To state them we first restrict to the valid zone. Since the logarithm is defined only on the positive side, its arguments must be positive.

Standing conditions

$$a \in (0,1) \text{ or } a > 1, \qquad M > 0, \quad N > 0, \qquad r \in \mathbb{R}.$$

The four laws

Law 1 (product): $$\log_a(MN) = \log_a M + \log_a N$$

Law 2 (quotient): $$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

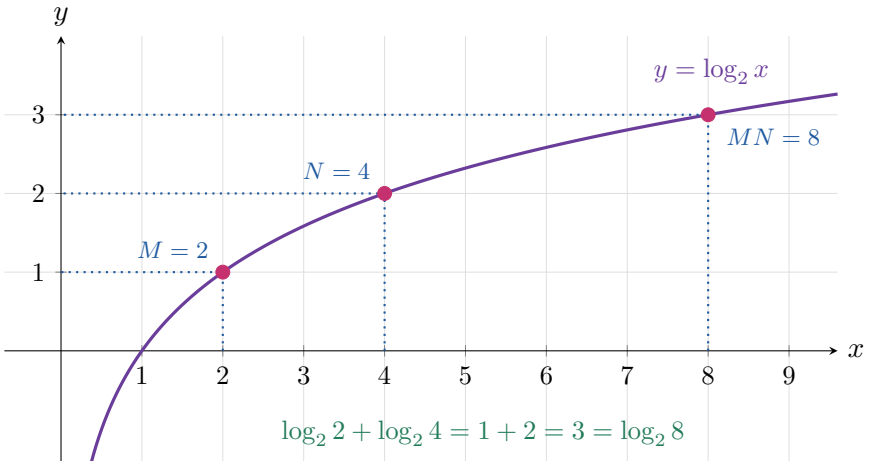
Law 3 (reciprocal): $$\log_a\left(\frac{1}{N}\right) = -\log_a N$$

Law 4 (power): $$\log_a\left(M^r\right) = r \log_a M$$

Stated verbally:

- The logarithm of a **product** of two positive numbers is the **sum** of the logarithms of the individual numbers.
- The logarithm of a **quotient** of two positive numbers is the **difference** of their logarithms.
- The logarithm of the **reciprocal** of a number is the **negative** of the logarithm of the original number. (This is not really a new law — we state it for clarity.)
- The logarithm of a positive number raised to a power r is r **times** the logarithm of that number.

The product law is visible on the graph itself: heights add up as arguments multiply.



4 Proof of Law 1 (product)

Put

$$A = \log_a M, \qquad B = \log_a N.$$

Now consider $A+B$, and raise a to that power. Since the exponential is the inverse of the logarithm,

$$a^{A+B} = a^{\log_a M + \log_a N} = a^{\log_a M} \cdot a^{\log_a N}.$$

But by the inverse property, $a^{\log_a M} = M$ and $a^{\log_a N} = N$. Therefore

$$a^{A+B} = M \cdot N.$$

Now take the logarithm to the base a on both sides:

$$\log_a \left(a^{A+B}\right) = \log_a(MN),$$

and since $\log_a (a^x) = x$, the left side is just $A + B$:

$$A + B = \log_a(MN).$$

Substituting back what A and B were,

$$\boxed{\log_a M + \log_a N = \log_a(MN)}$$

■

5 Proof of Law 2 (quotient)

You can now easily guess what modification is needed. Keep the same A and B; the only change is that we consider $A - B$ instead of $A + B$:

$$a^{A-B} = \frac{a^{\log_a M}}{a^{\log_a N}} = \frac{M}{N}.$$

Everything else changes correspondingly: instead of MN we obtain $\frac{M}{N}$. Taking the logarithm on both sides as before,

$$\log_a \left(a^{A-B}\right) = \log_a\left(\frac{M}{N}\right) \qquad \Longrightarrow \qquad A - B = \log_a\left(\frac{M}{N}\right),$$

and using the inverse property to cancel, we get

$$\boxed{\log_a M - \log_a N = \log_a\left(\frac{M}{N}\right)}$$

■

Practice

The rest of the steps are exactly the same as in Law 1 — work through them yourself as an exercise. The second law simply follows by replacing the plus sign with a minus sign.

6 Proof of Law 3 (reciprocal)

Simply apply the second law with $M = 1$:

$$\log_a\left(\frac{1}{N}\right) = \log_a 1 - \log_a N.$$

And what is $\log_a 1$? We have already seen that $(1, 0)$ lies on the graph of the logarithm, so $\log_a 1 = 0$. Hence

$$\boxed{\log_a\left(\frac{1}{N}\right) = -\log_a N}$$

■

7 Proof of Law 4 (power)

We apply the same modus operandi: first isolate a term, then look at it. Begin with the case

$$r \in \mathbb{N},$$

where, in our course, the set of natural numbers is

$$\mathbb{N} = \{0, 1, 2, \ldots\}.$$

If r is a natural number, the answer is easy. Write M^r as a repeated product:

$$\log_a\left(M^r\right) = \log_a\Big(\underbrace{M \cdot M \cdots M}_{r \text{ times}}\Big).$$

Now apply the multiplication rule — Law 1 — repeatedly:

$$= \underbrace{\log_a M + \log_a M + \cdots + \log_a M}_{r \text{ times}} = r \log_a M.$$

Therefore

$$\boxed{\log_a\left(M^r\right) = r \log_a M}$$

What about other r?

When r is a **rational** number the situation becomes tricky, though it can still be managed; we will not prove it here and will assume it for convenience. When r is a **real** number, the answer lies in calculus: one constructs a sequence of rationals converging to that real number — exactly the same logic we met when the number e arose for the natural exponential function. So what we have given here is a **partial proof**, valid for $r \in \mathbb{N}$;

8 Why Law 4 matters

For our purposes the important consequence is that even an irrational exponent comes out in front. For example,

$$\log_a\left(M^\pi\right) = \pi \log_a M.$$

Imagine an irrational number sitting in the index of a positive number: the law brings it down into a plain multiplication with respect to the logarithm. This simplifies calculations significantly, and this law will be used left and right in what follows.

Lecture 5 — Using the Laws of Logarithms

Expanding an expression; a warning about sums; combining terms into one logarithm

1 Putting the laws to work

Let us now use the laws of logarithms and look at some simple problems — how a problem can be *simplified* using logs, and equally how a problem can be made *compact* using logs.

2 Expanding a complicated expression

Example 5.1

Simplify, using the laws of logarithms,

$$\log_a\left(\frac{x^3\sqrt{x^2+1}}{(x+3)^4}\right).$$

It looks like an ugly product and quotient. The trick is to first **identify or isolate** the terms that can be separated. Here there are three such components:

$$x^3, \qquad \sqrt{x^2+1} = \left(x^2+1\right)^{1/2}, \qquad (x+3)^4.$$

So the end result after simplification should have three components.

Step 1: the quotient rule

Apply the quotient rule $\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$, which separates numerator from denominator:

$$\log_a\left(\frac{x^3\left(x^2+1\right)^{1/2}}{(x+3)^4}\right) = \log_a\left(x^3\left(x^2+1\right)^{1/2}\right) - \log_a\left((x+3)^4\right).$$

No other rule has been used so far — this is simply the second law.

Step 2: the product rule on the first term

The first term is a product of two factors, so by the first law,

$$\log_a\left(x^3\left(x^2+1\right)^{1/2}\right) = \log_a\left(x^3\right) + \log_a\left(\left(x^2+1\right)^{1/2}\right).$$

The exponent $\frac{1}{2}$ is written inside the logarithm to make clear that it sits on the argument.

Step 3: the power rule on each term

Now look at the last term: something is raised to the power 4. Do we have a rule for indices inside the argument of a logarithm? Yes — we have just proved it (for natural numbers). So

$$-\log_a\left((x+3)^4\right) = -4\log_a(x+3).$$

Doing the same for the other two terms gives the final answer:

$$\log_a\left(\frac{x^3\sqrt{x^2+1}}{(x+3)^4}\right) = 3\log_a x \;+\; \tfrac{1}{2}\log_a\!\left(x^2+1\right) \;-\; 4\log_a(x+3)$$

A remark on the $\frac{1}{2}$

For the middle term we are using the power rule with a **rational** exponent, which we have not proved — recall that we proved Law 4 only for natural numbers and assumed the rest.

Notice what has happened: the expression is now almost entirely made of **linear** terms, except for the one quadratic term inside a logarithm.

3 A warning: sums are not products

When studying logarithms, people generally get confused between a product becoming a sum. Terms such as $M+N$ inside a logarithm **cannot** be handled with the laws at all.

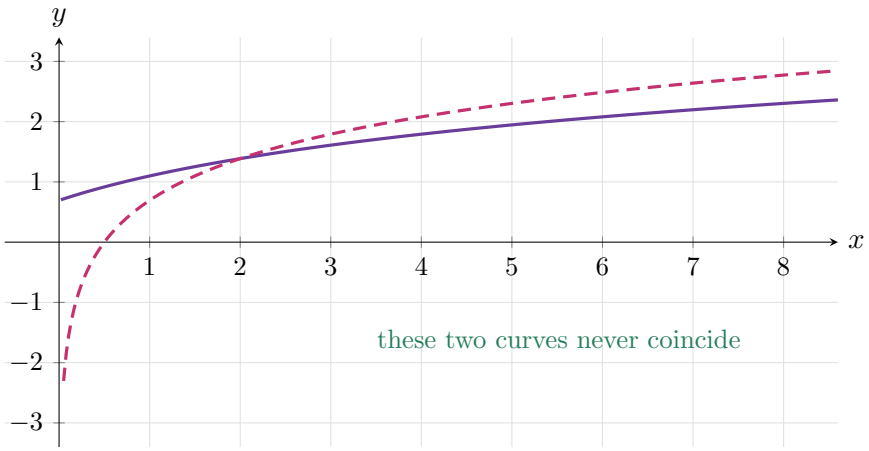
★ Warning — these are NOT equal ★

$$\log_a(M+N) \;\neq\; \log_a M + \log_a N, \qquad \text{because } \log_a M + \log_a N = \log_a(MN).$$

$$\log_a(M-N) \;\neq\; \log_a M - \log_a N, \qquad \text{because } \log_a M - \log_a N = \log_a\left(\frac{M}{N}\right).$$

Remember this warning, because generally when you are in the fighting spirit of solving a problem you tend to make exactly these mistakes — and they will ruin the entire answer. Nothing can be extracted from $\log_a(M+N)$: you cannot use a reverse exponential or any other form to get something out of it. Whenever such a term appears, be aware and do not apply the rules blindly.

The graph makes the difference plain: the two expressions are simply different functions.



$y = \ln(x+2)$ $y = \ln x + \ln 2 = \ln(2x)$

4 Combining terms into a single logarithm

The next question is the reverse one: **can we combine** using logs? If you do not want to look at such a big expression and would rather have a nice compact one, the answer is yes. Let us handle the terms one by one and merge them.

Example 5.2

Combine into a single logarithm:

$$2\log_a x + \log_a 9 + \log_a\left(x^2+1\right) - \log_a 5.$$

Step 1: the first two terms

We know $\log_a\left(x^r\right) = r\log_a x$. Apply it **in reverse**:

$$2\log_a x = \log_a\left(x^2\right).$$

Do not stop there — now apply the product rule to merge it with $\log_a 9$:

$$\log_a\left(x^2\right) + \log_a 9 = \log_a\left(9x^2\right).$$

Step 2: the next two terms

The remaining two terms carry a negative sign between them, so naturally the quotient rule applies:

$$\log_a\left(x^2+1\right) - \log_a 5 = \log_a\left(\frac{x^2+1}{5}\right).$$

Step 3: combine the two results

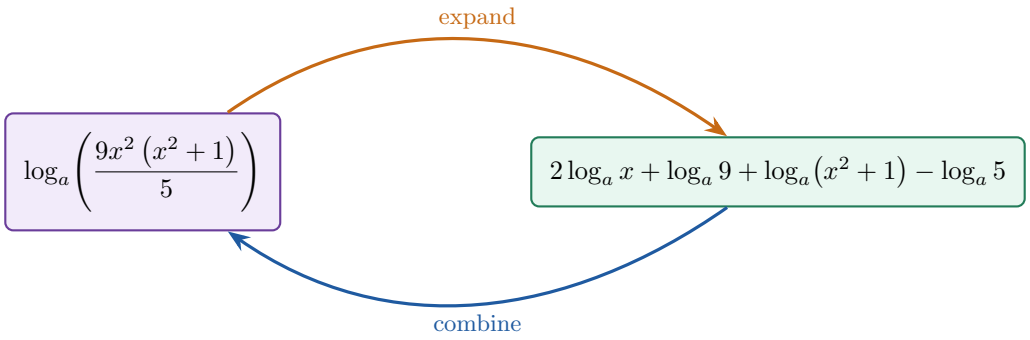
Can we combine these two? Again apply the product rule:

$$\boxed{2\log_a x + \log_a 9 + \log_a\left(x^2+1\right) - \log_a 5 = \log_a\left(\frac{9x^2\left(x^2+1\right)}{5}\right)}$$

5 Which direction to go?

Both directions are useful, and they help in opposite situations.

Direction	When it helps
Expand into a long sum of terms	When there are many extremely **large** numbers to be calculated
Combine into one compact logarithm	When many **small** terms are unnecessarily occupying space



So these are the two simple avenues available: when extremely large terms appear, expanding simplifies the calculation; when many small terms clutter the page, combining tidies it up.

Lecture 6 — The Change of Base Rule

Why only two logarithms are needed; statement, proof and applications

1 A fact worth revising

We already know some facts, but it is better to revise them while actually handling problems. Here is one such result.

Theorem 6.1

For $a \in (0,1)$ or $a > 1$, and $M, N > 0$:

$$\log_a M = \log_a N \qquad \Longrightarrow \qquad M = N.$$

The conditions here are simply taking care of the fact that we stay in the valid domain of the logarithm.

2 The two logarithms we actually use

When solving any engineering problem — or anything involving an engineering discipline — the base a is typically restricted to just two numbers: e (Euler's constant) and 10. These carry special names and special notation.

Name	Base	Notation	Calculator key
Natural logarithm	e	$\ln x$	`ln`
Common logarithm	10	$\log x$	`log`

The **natural logarithm** arises naturally in the theory of calculus, and therefore has special value; on a scientific calculator there is a specific provision for it, marked `ln`. The **common logarithm** represents our decimal system — in the decimal system everything is given in powers of 10 — and since we commonly use the decimal system, it is called the common logarithm. On a calculator it appears simply as `log`.

These are the two important logarithms we study.

3 Why only these two?

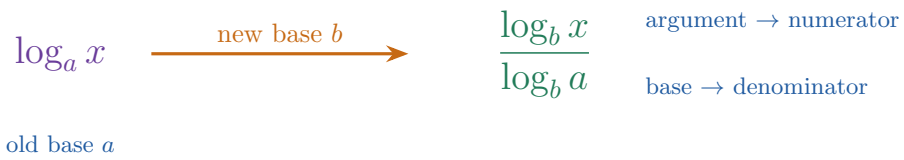
The natural question is: why only these two, and not others? The bold claim is that **these two logarithms suffice** for studying logarithms entirely — and the answer to *why* is given by the following theorem.

Theorem 6.2 — Change of base rule

For $x > 0$, with the **old base** $a \in (0,1)$ or $a > 1$, and the **new base** $b \in (0,1)$ or $b > 1$:

$$\boxed{\log_a x = \frac{\log_b x}{\log_b a}}$$

The recipe is easy to remember: choose whichever new base you want, then write the **argument** of the function in the numerator and the **base** in the denominator, taking logarithms to the new base in both places.



So if the theorem holds, then you may hand me a logarithm to *any* base and I will convert it into a logarithm to the base 10 or to the base e and compute accordingly. That is the beauty of this theorem, and it is exactly why we have only two important logarithms. It is also why scientific calculators are designed with only the two keys `ln` and `log`, and generally offer no arbitrary base. (An advanced calculator may allow any base, but these two suffice in general.)

The change of base is just scaling

What we are really saying is that a change of base is nothing but a **multiplication by a constant**, namely

$$\frac{1}{\log_b a}.$$

So the graph of $\log_a x$ is just a vertically scaled copy of the graph of $\log_b x$.

4 Proof of the change of base rule

The proof is easy once you understand the key logic behind all the laws of logarithms: the **exponential function is the inverse of the logarithmic function**. That is all we use.

Put

$$M = \log_a x, \qquad N = \log_b x, \qquad R = \log_b a.$$

Recall the other description of a logarithm: asking for $\log_a x$ is asking *to what power must the base a be raised in order to get x?* — and the answer is M. So applying the exponential,

$$a^M = a^{\log_a x} = x.$$

Using the same reasoning for the other two terms,

$$b^N = x, \qquad b^R = a.$$

Now, since a can be written as b^R, substitute it into the first expression:

$$a^M = \left(b^R\right)^M = x.$$

Simplifying the indices,

$$b^{RM} = x.$$

Hit both sides with the logarithm to the base b:

$$\log_b \left(b^{RM}\right) = \log_b x,$$

and since the logarithm is the inverse of the exponential, the left side collapses:

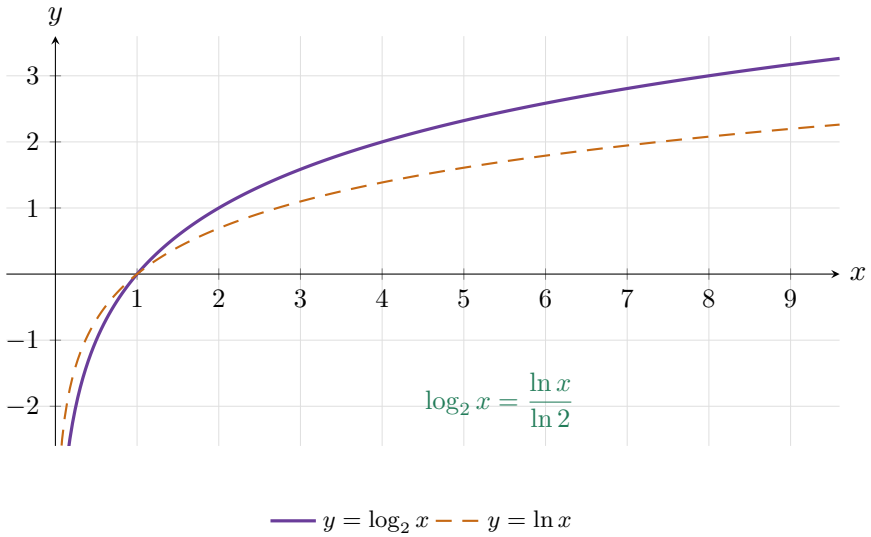
$$RM = \log_b x.$$

Finally substitute back what R and M were:

$$\log_b a \cdot \log_a x = \log_b x \qquad \Longleftrightarrow \qquad \log_a x = \frac{\log_b x}{\log_b a}.$$

■

So switching from base a to base b is simply a multiplication by the constant $\log_b a$ — which is what we claimed. Henceforth you may forget about all other bases and work only with the natural or the common logarithm; that is the key idea we will follow.



5 Applying the rule

Example 6.1

Find $\log_5 89$.

You need not worry about what a logarithm to the base 5 means in practice. Simply use the natural logarithm together with the change of base formula:

$$\log_5 89 = \frac{\ln 89}{\ln 5} = 2.7889\ldots \approx 2.79.$$

Your calculator has an `ln` key, so compute it directly.

Example 6.2

Find $\log_{\sqrt{2}} \sqrt{5}$.

Somebody may hand you an absurd-looking base — even an irrational one. Still there is nothing to worry about: apply the change of base formula, sending the argument to the numerator and the base to the denominator,

$$\log_{\sqrt{2}} \sqrt{5} = \frac{\ln \sqrt{5}}{\ln \sqrt{2}} = 2.32.$$

So no matter what base is given to you, the calculation simplifies and all such problems can be solved easily with a scientific calculator.

6 A warning

Sometimes people confuse the change of base formula with the following, writing

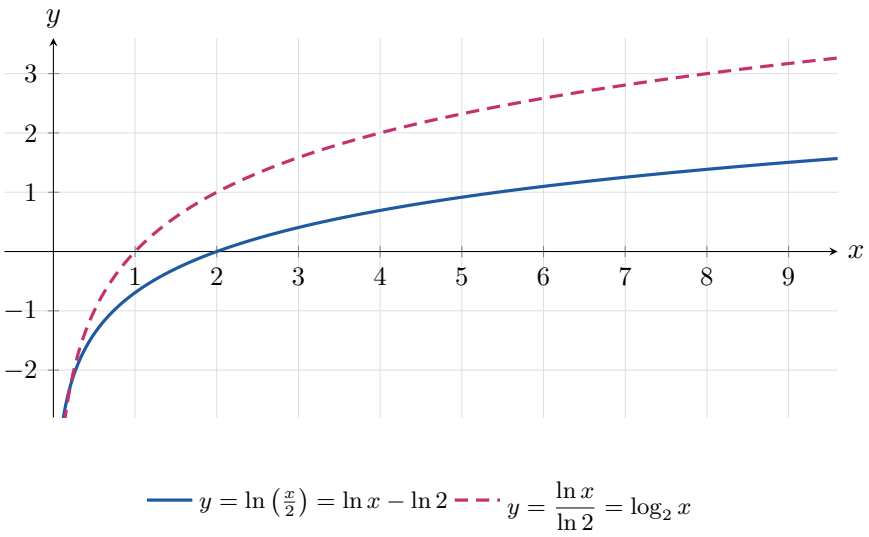
$$\ln\left(\frac{x}{a}\right) \quad \text{as} \quad \frac{\ln x}{\ln a}.$$

★ Warning — this identity is NOT true ★

$$\ln\left(\frac{x}{a}\right) \neq \frac{\ln x}{\ln a}$$

The left side is $\ln x - \ln a$ by the quotient law; the right side is $\log_a x$ by the change of base rule. They are entirely different quantities.

So be careful while solving problems: in the verge of solving, you may tend to make exactly this kind of mistake, which will ruin the entire answer.



In the next lecture we will look at better ways of graphing the logarithmic function to any base using natural logarithms, along with some other variations.

Lecture 7 — Solving Logarithmic Equations

Graphing by change of base; an inequality; equations with same and different bases

1 Graphing a logarithm to any base

Example 7.1

Plot the graph of

$$f(x) = \log_2 x.$$

Suppose you have a graphing calculator. Older versions often do not allow a logarithm to the base 2 to be taken — all you have available is

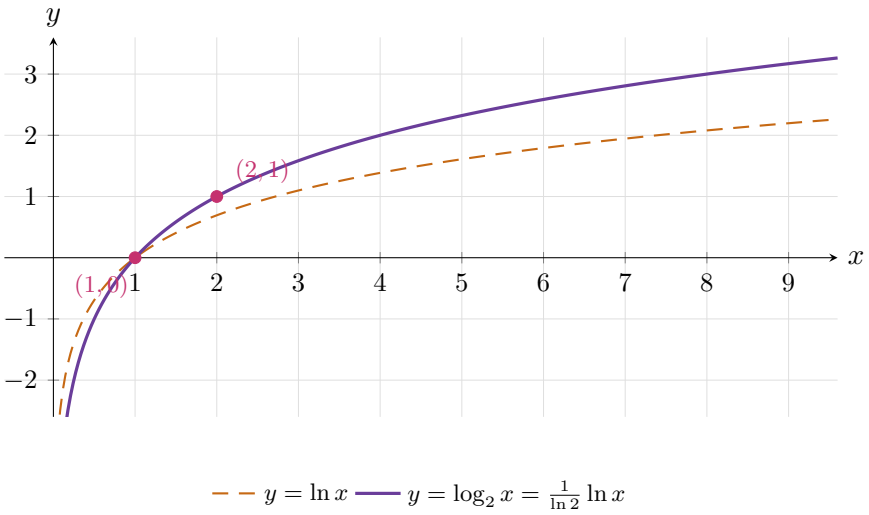
$$\log_{10} x \qquad \text{or} \qquad \log_e x = \ln x.$$

Only these two are available, and both can be plotted. Can we then plot $\log_2 x$?

Using the **change of base formula** from the last lecture, convert the function to base e (for convenience):

$$f(x) = \log_2 x = \frac{\ln x}{\ln 2} = \frac{1}{\ln 2} \cdot \ln x.$$

That simply means: plot $\ln x$ and **scale it** by the constant $\frac{1}{\ln 2}$, whose value any calculator will give. So the graph has more or less the same features as $\ln x$ — only appropriately scaled.



$y = \ln x$ $y = \log_2 x = \frac{1}{\ln 2} \ln x$

This is a clear-cut demonstration of the usability of the change of base formula: you need no longer bother about which base the function is given to — simply convert it to base e or base 10.

2 Proving an inequality

Example 7.2

Prove that

$$\frac{1}{\log_2 \pi} + \frac{1}{\log_6 \pi} > 2.$$

The key tool

While solving *any* logarithmic equation or inequality, the key tool is **exponentiation**. We use it throughout below.

Here one base is 2 and the other is 6 — we do not want that. We want everything to the same base, so we again use the change of base formula. Take the left-hand side:

$$\text{LHS} = \frac{1}{\log_2 \pi} + \frac{1}{\log_6 \pi}.$$

Now $\log_2 \pi = \dfrac{\ln \pi}{\ln 2}$, and since it sits in a denominator, its reciprocal is

$$\frac{1}{\log_2 \pi} = \frac{\ln 2}{\ln \pi}.$$

In a similar manner,

$$\frac{1}{\log_6 \pi} = \frac{\ln 6}{\ln \pi}.$$

Something amazing has happened: the denominators are now the same. So

$$\text{LHS} = \frac{\ln 2 + \ln 6}{\ln \pi}.$$

Do we know anything about $\log a + \log b$? Yes — it is $\log(ab)$. Hence

$$\text{LHS} = \frac{\ln(2 \times 6)}{\ln \pi} = \frac{\ln 12}{\ln \pi}.$$

Now the question is whether this exceeds 2. We do not yet know, so let us carry the question along:

$$\frac{\ln 12}{\ln \pi} \overset{?}{>} 2.$$

We know $\ln \pi$ is non-zero, and it is **positive**, so we may multiply throughout by it without disturbing the inequality:

$$\ln 12 \overset{?}{>} 2 \ln \pi.$$

Using the power rule $\log(x^a) = a \log x$ in reverse, $2 \ln \pi = \ln(\pi^2)$:

$$\ln 12 \overset{?}{>} \ln(\pi^2).$$

Both sides are now logarithms to the same base, so we may **exponentiate** with e. Since the exponential function is monotonically increasing, the inequality stays intact:

$$e^{\ln 12} \overset{?}{>} e^{\ln(\pi^2)}.$$

By the inverse-function definition, $a^{\log_a x} = x$, so this reduces to

$$12 \overset{?}{>} \pi^2.$$

Now it is easy. What is π? It is $3.141\ldots$, which is strictly smaller than 3.15. Therefore

$$\pi^2 < (3.15)^2 < 10 < 12.$$

So the inequality under question is certainly true, every question mark may be dropped, and hence

$$\boxed{\frac{1}{\log_2 \pi} + \frac{1}{\log_6 \pi} > 2}$$

The techniques used

Three things were used together: the **change of base** formula, the **multiplication (product) rule** — twice — and the **definition of the inverse function** (which is not a law, but the definition itself).

3 An equation with base less than one

Example 7.3

Solve for x:

$$2\log_{0.5} x = \log_{0.5} 4.$$

At the beginning you may worry about that 0.5, since for a base $0 < a < 1$ the behaviour of the log function was somewhat different. Do we really need to worry? Before worrying, let us simplify.

Apply the **power rule** to the left-hand side:

$$2\log_{0.5} x = \log_{0.5}\left(x^2\right),$$

so the equation becomes

$$\log_{0.5}\left(x^2\right) = \log_{0.5} 4.$$

The base is common on both sides, so the exponentiation trick $a^{\log_a x} = x$ works. Exponentiate with respect to 0.5:

$$0.5^{\log_{0.5}(x^2)} = 0.5^{\log_{0.5} 4} \qquad \Longrightarrow \qquad x^2 = 4.$$

From our knowledge of quadratic functions, $x^2 - 4 = 0$ has two roots:

$$x = +2 \quad \text{and} \quad x = -2.$$

Check feasibility

Are both solutions feasible in the original equation? Consider $\log_{0.5}(-2)$. The logarithmic function is defined only on the **positive** real line, so $\log_{0.5}(-2)$ is undefined — it lies outside the domain. That value is chucked off.

Therefore

$$\boxed{x = 2}$$

Substituting back: $2^2 = 4$, so $\log_{0.5} 4 = \log_{0.5} 4$. ✓

4 An equation needing the product rule first

Example 7.4

Solve for x:

$$\log_8(x+1) + \log_8(x-1) = 1.$$

It is good practice to write the equation out once as it is — doing so helps you see the niceties of the equation. Apply the multiplication rule $\log M + \log N = \log(MN)$:

$$\log_8\big((x+1)(x-1)\big) = 1.$$

Now we want to get rid of the log factor, so we **exponentiate** with the base 8, writing $(x+1)(x-1) = x^2 - 1$:

$$8^{\log_8(x^2-1)} = 8^1.$$

Do not skip this step

If you do not write the right-hand side as 8^1, you may mistakenly leave it as 1. Write all these steps out.

The 8 and $\log_8$ vanish against each other, leaving

$$x^2 - 1 = 8 \qquad \Longrightarrow \qquad x^2 - 9 = 0 \qquad \Longrightarrow \qquad x = \pm 3.$$

Now perform the same feasibility check as before:

- $x = +3$: the arguments become $3+1 = 4$ and $3-1 = 2$, both positive — valid, in the domain.
- $x = -3$: the arguments become $-3+1 = -2$ and $-3-1 = -4$, both negative — not permissible inside a logarithm, so -3 cannot be in the domain.

Hence

$$\boxed{x = 3}$$

Verification: $\log_8(4) + \log_8(2) = \log_8(8) = 1$. ✓

5 An equation with different bases

Example 7.5

Solve for x:

$$\log_3 x + \log_4 x = 4.$$

Here the exponentiation trick cannot be used directly, because the **bases are different**. The first thing to do is to make the bases equal — and the formula for that is the change of base formula. Writing the expression once more and converting to the natural base:

$$\frac{\ln x}{\ln 3} + \frac{\ln x}{\ln 4} = 4.$$

Take $\ln x$ common:

$$\ln x \left[\frac{1}{\ln 3} + \frac{1}{\ln 4}\right] = 4.$$

The quantity inside the square brackets is a non-zero number, so we may take it to the other side — where it appears as a reciprocal:

$$\ln x = \frac{4}{\dfrac{1}{\ln 3} + \dfrac{1}{\ln 4}}.$$

Simplifying, the denominators combine to put $\ln 3 + \ln 4$ below and $\ln 3 \cdot \ln 4$ above:

$$\ln x = \frac{4 \ln 3 \ln 4}{\ln 3 + \ln 4}.$$

Note that $\ln 3 \cdot \ln 4$ *cannot* be combined into anything — it stays as it is. But the denominator can, by the multiplication rule: $\ln 3 + \ln 4 = \ln 12$. So

$$\ln x = \frac{4 \ln 3 \ln 4}{\ln 12}.$$

Since this is a natural logarithm, exponentiate:

$$\boxed{x = e^{\left(\frac{4 \ln 3 \ln 4}{\ln 12}\right)}}$$

If you wish, you may merge the 4 with one of the logarithms and write $\ln\left(3^4\right)$ or $\ln\left(4^4\right)$, whichever you are comfortable with; we leave it as it is. This is a perfectly valid solution, and this is how a problem with different bases is solved.

6 An equation that looks ambiguous

Example 7.6

Solve for x:

$$\ln\left(x^2\right) = (\ln x)^2 .$$

Because of the square, some knowledge of quadratic functions is needed here. First ask: are the bases common? Yes — both are natural logarithms, so there is no problem there.

The difficulty is that one side involves $\ln\left(x^2\right)$ and the other $\ln x$. Can we bring both into the form $\ln x$? The power law comes to our aid:

$$\ln\left(a^k\right) = k \ln a \qquad \Longrightarrow \qquad 2 \ln x = (\ln x)^2 .$$

Now comes the real strength of the method: both terms have the same argument, so we may treat this as a **composition** and substitute. Put

$$t = \ln x \qquad \Longrightarrow \qquad 2t = t^2.$$

Then

$$2t - t^2 = 0 \qquad \Longrightarrow \qquad t\,(2 - t) = 0 \qquad \Longrightarrow \qquad t = 0 \quad \text{or} \quad t = 2.$$

But what is t? By our substitution it is $\ln x$, so we are asking:

$$\ln x = 0 \qquad \text{or} \qquad \ln x = 2.$$

In both cases we have a natural logarithm, so exponentiate:

$$e^{\ln x} = e^0 \quad \Longrightarrow \quad x = 1, \qquad\qquad e^{\ln x} = e^2 \quad \Longrightarrow \quad x = e^2.$$

Hence

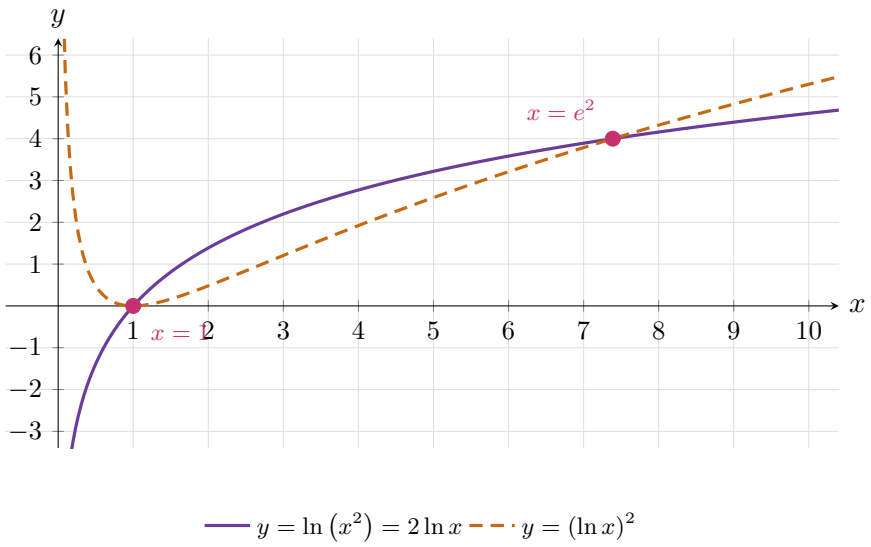
$$\boxed{x = 1 \quad \text{or} \quad x = e^2}$$

Verification

It is good to go back to the original problem and check both.

- $x = 1$: then $\ln\left(1^2\right) = \ln 1 = 0$, and $(\ln 1)^2 = 0^2 = 0$. ✓
- $x = e^2$: then $\ln\left(e^4\right) = 4$, while $\ln\left(e^2\right) = 2$ and $2^2 = 4$. ✓

Both are valid answers. It is worth plotting the two curves with a graphing tool such as Desmos and seeing where they intersect — the points of intersection are exactly the solutions we have found.



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